

# Privacy-preserving Resilient Consensus for Multi-agent Systems in a General Topology Structure

JIAN HOU and JING WANG, Zhejiang Sci-Tech University, China  
MINGYUE ZHANG and ZHI JIN, Peking University, China  
CHUNLIN WEI, Beijing Institute of Control Engineering, China  
ZUOHUA DING, Zhejiang Sci-Tech University, China

Recent advances of consensus control have made it significant in multi-agent systems such as in distributed machine learning, distributed multi-vehicle cooperative systems. However, during its application it is crucial to achieve resilience and privacy; specifically, when there are adversary/faulty nodes in a general topology structure, normal agents can also reach consensus while keeping their actual states unobserved.

In this article, we modify the state-of-the-art Q-consensus algorithm by introducing predefined noise or well-designed cryptography to guarantee the privacy of each agent state. In the former case, we add specified noise on agent state before it is transmitted to the neighbors and then gradually decrease the value of noise so the exact agent state cannot be evaluated. In the latter one, the Paillier cryptosystem is applied for reconstructing reward function in two consecutive interactions between each pair of neighboring agents. Therefore, multi-agent privacy-preserving resilient consensus (MAPPRC) can be achieved in a general topology structure. Moreover, in the modified version, we reconstruct reward function and credibility function so both convergence rate and stability of the system are improved.

The simulation results indicate the algorithms' tolerance for constant and/or persistent faulty agents as well as their protection of privacy. Compared with the previous studies that consider both resilience and privacy-preserving requirements, the proposed algorithms in this article greatly relax the topological conditions. At the end of the article, to verify the effectiveness of the proposed algorithms, we conduct two sets of experiments, i.e., a smart-car hardware platform consisting of four vehicles and a distributed machine learning platform containing 10 workers and a server.

CCS Concepts: • **Security and privacy** → **Network security**;

Additional Key Words and Phrases: Resilient consensus, multi-agent systems, privacy preserving, Q-consensus

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Authors' addresses: J. Hou, J. Wang, and Z. Ding, Zhejiang Sci-Tech University, Hangzhou Shi, Zhejiang Province, China; emails: {changeleap, m18656897772}@163.com, zouhuading@hotmail.com; M. Zhang (corresponding author) and Z. Jin, Peking University, Haidian Qu, Beijing Shi, China; emails: {mingyuezhang, zhijin}@pku.edu.cn; C. Wei, Beijing Institute of Control Engineering, Haidian Qu, Beijing Shi, China; email: clwei502@163.com.

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**1 INTRODUCTION**

Over the past few years, **Distributed Artificial Intelligence (DAI)**, which is concerned with solving the problem by a decentralized group of agents<sup>1</sup> working cooperatively, has gained much attention from academia to industry [4, 28, 33]. With the advancing technology, improving requirement of system quality, and the interrupting running environments that are unpredictable and possibly hostile, three fundamental challenges have thus naturally arisen: (i) *the fact that some agents might behave abnormally subject to external attack or internal faults*, (ii) *individual information should adhere to certain privacy-preserving requirement*, and (iii) *the communication topology is unknown and uncertain*.

Recent progress in DAI has witnessed significant applications regarding the challenges above. The privacy-preserving distributed machine learning is a trending application of this kind, in which private data is partitioned into one or multiple servers and several workers to train a common model without privacy disclosure. A typical scenario is disease diagnosis in which medical institutions share the encrypted information of their patients (e.g., medical history, lab results) to develop machine learning models that can accurately diagnose disease [20, 27]. In this case, all the servers and workers interact with each other by gradients or model parameters to ensure privacy [36], while some faulty servers or workers may provide malicious data that will cause disturbance to the system. Another instance is that multiple vehicles cooperate in a distributed manner to achieve a certain goal such as rendezvous or formation in the presence of uncontrollable vehicles and with the privacy of each state protected. Apart from the examples above, Blockchain and clock synchronization may also confront such challenges.

In essence, the model parameters of both servers and workers in distributed machine learning will gradually achieve consensus, so will relative state between agents in multi-vehicle systems. Such consensus of key parameters should also be achieved in Blockchain, clock synchronization, and so on (as shown in Figure 1). Therefore, **multi-agent privacy-preserving resilient consensus (MAPPRC)** in a general topology structure, i.e., reaching a global agreement with only local but possibly false data injection interactions subject to the privacy-preserving requirement, is one of the fundamental concerns [7, 38].

In conventional approaches, resilient consensus employs the explicit exchange of state values among neighboring nodes to abandon some extreme values, thereupon, as long as the strict topology condition [35, 37] is satisfied, the normal agents are able to achieve resilient consensus regardless of the behavior of faulty agents. In practical applications, unfortunately, the topology condition is commonly tough and unrealizable. Some works, therefore, expect to evaluate and identify the status of neighboring agents through the analysis of historical data, however, it brings along storage and computation loads at the same time. On the other aspect, both noise and cryptography are considered to guarantee the privacy-preserving requirement of each node. Generally speaking, cryptography-based approach could converge in a deterministic manner, yet noise-based approach takes the advantage over low computation and communication loads. Some recent works begin to take resilience and privacy-preserving requirement into account at the same time [6, 29], which are

<sup>1</sup>In some literature, agent is also referred to as node.

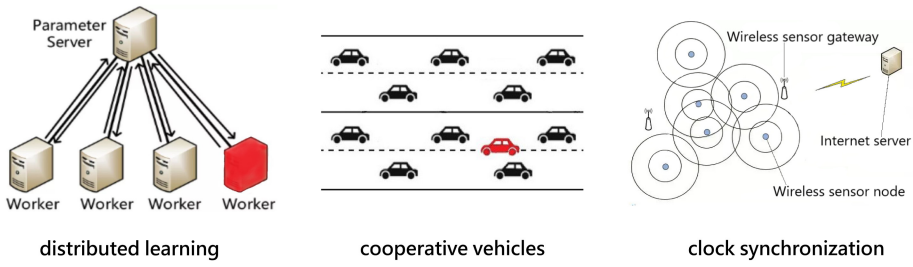


Fig. 1. Some DAI applications under resilience and privacy-preserving requirements.

also subject to the strict topology conditions above. Nevertheless, MAPPRC in a general topology structure is still an open issue.

As far as we know, Q-consensus algorithm [13] is a closely related field on solving resilient consensus problem in a general topology structure. However, MAPPRC presents two challenges from Q-consensus algorithm perspective: (i) *Q-consensus algorithm takes accurate (relative) state information as the control input, which is completely unrealizable in MAPPRC for its privacy requirement for each node*, (ii) *it is of weak stability, i.e., system can hardly achieve accurate consensus, as the weights of faulty agents may not be reduced to 0*.

This article introduces noise/cryptography and redesigns Q-consensus algorithm to overcome these challenges for achieving MAPPRC in distributed systems. The main idea underlying our algorithm is that through establishing reward function and credibility function for each pair of neighboring agents, the adjacent weights can update in an adaptive way. Specifically, on the one hand, to ensure each agent's privacy, we either endow each agent state with a specified noise that follows independent zero-mean random variable with Laplace distribution and variance or introduce a Paillier cryptosystem to reconstruct the reward function in two consecutive interactions that will also improve the algorithm's evaluating accuracy. On the other hand, we rebuild reward function by bringing previous impact to the current assessing process via corresponding adjacent weight, as well as credibility function by introducing horizontal assessments to improve the system's convergence rate.

Two types of faulty agents, i.e., persistent faulty agent and constant faulty agent, both with certain specified percentage, are considered. Last, we conduct experiments in three ways, i.e., numerical simulations, smart-car resilient rendezvous, and distributed resilient learning, to evaluate the effectiveness of the proposed algorithm. The main contributions are listed as follows:

- (i) Compared with previous studies, normal agents in the proposed approach are able to achieve resilient consensus in the presence of faulty agents subject to privacy-preserving requirements, i.e., MAPPRC, in any general topology structure, which has not been implemented before.
- (ii) Compared with the previous version, the modified Q-consensus algorithm boasts better performance, including the improvement of convergence rate and system stability, thanks to the reconstruction of reward function and credibility function.
- (iii) The proposed algorithms are applied to experiments in three ways, i.e., numerical simulations, smart-car resilient rendezvous, and distributed resilient learning, all of which demonstrate their potential in future studies.

The rest of this article is organized as follows: Section 2 presents related work. Section 3 describes the problem formulation. Sections 4 and 5 give the Noise-based MAPPRC algorithm and Cryptography-based MAPPRC algorithm, respectively. Section 6 conducts the experiments and presents the results. Section 7 gives a short conclusion and outlines future work.

## 2 RELATED WORK

In DAI, system-security has been emphasized for a long period, with the purpose against malicious entities. Among all issues, multi-agent consensus, with only local interaction among agents, plays a fundamental role aiming at achieving global agreement in an appropriate sense. The features of consensus make it widely applied in various fields, including distributed optimization [28], clock synchronization, formation control [4], Pagerank [14] and so on.

### 2.1 Resilient Consensus

A relevant property in consensus problem is its robustness on faults, i.e., how to achieve consensus in the presence of faulty nodes. Typically, the functional method is to remove suspicious nodes at state updating steps.

Supposing that the maximum number  $F$  of faulty agents is known *a priori*, each agent abandons some extreme values among the received information from the neighboring agents. Regularly, such scheme is called **Mean-Subsequence-Reduced (MSR)** algorithm [17] and then followed the **Weighted-Mean-Subsequence-Reduced (W-MSR)** algorithm [18]. In Reference [9], Haseltalab et al. presented a fault-tolerant algorithm in asynchronous networks, and the corresponding convergence rate and time analysis were given in their following work [10]. In Reference [3], a resilient consensus problem with double-integrator dynamics was then studied, in which the faulty nodes were aware of the updating times and time-delay. In Reference [34], each node in the system was equipped with an independent clock and thus implemented state updating at arbitrary instants in event-triggered and self-triggered control. In Reference [30], a general algorithm was proposed for a wide range of consensus; for this, not only could faulty nodes be identified, but also the converged value were kept within the initial values of normal nodes. In addition, the resilient consensus algorithm was also applied in resilient formation control for mobile robots in the presence of noncooperative robots [33].

Fundamentally, the MSR-type algorithm is skilled in convergence and robustness, however, it is constrained by network connectivity [35] or network robustness [37], resulting in disabled implementation in general networks. Besides, Reference [25] provided an algorithm so-called RoboTrust, evaluating the trustworthiness for each agent by observations and statistical inferences from various historical perspectives. Apart from three decision rules by local trust evaluation, Baras et al. in Reference [1] further introduced a global trust evaluation so the robustness was enhanced.

To achieve resilient consensus in general graph structures with less communication and computation loads, Q-consensus algorithm was recently presented in Reference [13]. The whole evaluating process of Q-consensus will be realized through alternating horizontal and vertical assessment. Horizontal assessment will be conducted on each neighbor node, in which its immediate reward will be calculated, reflecting its immediate reliability. Vertical assessment refers to the comprehensive assessment on the reliability of each neighbor node, in which a new formula is developed to process the data we collected in horizontal assessments to work out the credibility of each neighbor node. Notably, adjustments will be made on the criteria of every horizontal assessment in the light of the previous vertical assessment. Finally, the adjacent weight for a neighbor node is defined as the proportion of that node's credibility to the total credibility of neighbor nodes, and the proposed assessing mechanism aims to minimize the corresponding weights of the abnormal nodes to zero, eliminating their influence to achieve consensus among all normal nodes. Consequently, as long as one of the neighbor nodes is normal, this algorithm will function properly. The specific instruction is referred to in Figure 2.

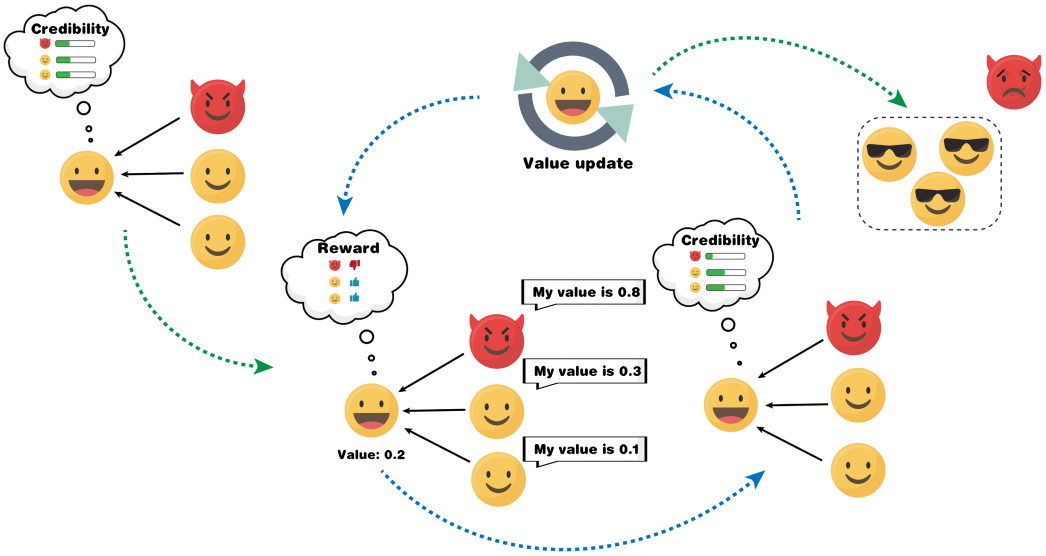


Fig. 2. Q-consensus algorithm.

## 2.2 Privacy-preserving Consensus

Another relevant property in consensus problem is its privacy-preserving requirement on agent state for the sake of security. Generally speaking, there are two most commonly used approaches, i.e., noise-based approach and cryptography-based approach, in light of privacy-preserving consensus problem.

In noise-based approach, nodes exchange information with random or predefined noises on their states. Reference [24] introduced pseudo-random offsets to achieve the average consensus asymptotically on the initial states of normal nodes. The mean square convergence rate and the covariance matrix of the maximum likelihood estimates were derived in Reference [26]. A noise-based mechanism was considered in Reference [15] on quadratic optimization problem, and a concept named “cooperation level” was introduced to balance the level between cooperation and privacy. Reference [12] provided a privacy definition to describe the inner relationship between the maximum disclosure probability and the estimation accuracy.

In general, the noise-based approach achieves average consensus in a statistic case, however, fails to obtain ideal performance in deterministic scenarios. Alternatively, the cryptography-based approach is introduced to guarantee convergence in a deterministic manner. Reference [16] provided a consensus gossip algorithm to exchange encrypted data using secure multi-party computation methodology, in which the nodes had no opportunity to access to the data directly when implementing iterations. In Reference [8], Hadjicostis et al. put forward a concept so-called ratio consensus and utilized homomorphic encryption strategy to achieve average consensus in the presence of a trusted node. Reference [5] further extended a secure and privacy-preserving consensus algorithm to second-order systems based on Paillier encryption.

## 2.3 Privacy-preserving Resilient Consensus

The first literature striking a balance between resilient consensus and privacy-preserving requirement, i.e., MAPPRC, was referred to in Reference [7], in which the agents utilized **Differentially Private MSR (DP-MSR)** algorithm to reach global agreement subject to a differential privacy requirement and the interruption of certain faulty agents, and the corresponding accuracy

Table 1. Comparisons of Different Algorithms

| Algorithm             | Reference                         | Resilient Consensus | Privacy-preserving | Necessary conditions/characteristics    |
|-----------------------|-----------------------------------|---------------------|--------------------|-----------------------------------------|
| 1. MSR-type           | [3, 9, 10, 17, 18, 30, 33–35, 37] | ✓                   | ×                  | network connectivity/network robustness |
| 2. RoboTrust          | [25]                              | ✓                   | ×                  | strong computing capability             |
| 3. Noise-based        | [12, 15, 24, 26]                  | ×                   | ✓                  | lower convergence rate                  |
| 4. Cryptography-based | [5, 8, 16]                        | ×                   | ✓                  | strong computing capability             |
| 5. DP-MSR             | [6, 7, 29]                        | ✓                   | ✓                  | network connectivity/network robustness |
| 6. Q-consensus        | [13]                              | ✓                   | ×                  | weak stability                          |

property was further discussed in their following work [6]. In addition, Reference [29] employed the cryptography of homomorphic encryption to protect the privacy of the initial states and in the meanwhile, to achieve resilient average consensus. A comparison of different algorithms is shown in Table 1.

In conclusion, the existing works concentrate more on certain topology structure, and thus, a MAPPRC algorithm in a general topology structure is of significance.

### 3 PRELIMINARIES AND PROBLEM FORMULATION

In this section, we will briefly review graph theory and give our system model of this article.

#### 3.1 Graph Theory

We first give the following definition on a multi-agent system:

*Definition 3.1.* A multi-agent system is defined as a 5-tuple  $\mathcal{M} = \langle V, E, X, \bar{X}, U \rangle$ , where:

- $V$  is a finite set of  $n$  agents labeled by  $1, 2, \dots, n$ .
- $E \subseteq V \times V$  is a finite set of edges that describes the neighbor relations.
- $X \subseteq \mathbb{R}$  is a set of actual states.  $x_i(k)$  denotes the actual state of agent  $i$  at time  $k$ .  $\mathbf{x}(k) = [x_1(k), \dots, x_n(k)]^T$  is the joint state of all agents at time  $k$ .
- $\bar{X} \subseteq \mathbb{R}$  is a set of observation states.  $\bar{x}_i(k)$  denotes the interacted information of agent  $i$  at time  $k$ .
- $U \subseteq \mathbb{R}$  is a set of control inputs.  $u_i(k)$  indicates the control input of agent  $i$  at time  $k$ .

Specifically, node  $j$  is called a neighbor of node  $i$ , with an edge from  $j$  to  $i$ , only if node  $i$  can receive information  $\bar{x}_j$  from node  $j$ . We write the neighbor set of node  $i$  by  $N_i = \{j \mid j \neq i, (j, i) \in E\}$ . In this article, we consider a fixed topology, i.e.,  $N_i = N_i(k)$ . A path from node  $j$  to node  $i$  is a sequence of distinct nodes  $i_0, i_1, \dots, i_m$ , where  $i_0 = j$  and  $i_m = i$  with  $(i_l, i_{l+1}) \in E, 0 \leq l \leq m-1$ . We say a directed graph  $G = (V, E)$  rooted if and only if there exists  $i \in V$ , so there is a path from node  $i$  to any other node.

Formally, let  $V' \subset V$  be any subset of  $V$ . Then, *induced subgraph*  $G[V'] = \langle V', E' \rangle$  is the graph satisfying that  $E' = \{(i, j) \mid i \in V', j \in V', (i, j) \in E\}$ . And  $G - G[V'] = \langle V'', E'' \rangle$  denotes the graph satisfying that  $V'' = \{i \mid i \in V, i \notin V'\}$  and  $E'' = \{(i, j) \mid (i, j) \in E, (i, j) \notin E'\}$ . We call the set of normal nodes  $V^n$ , the set of persistent faulty nodes  $V^p$ , and the set of constant faulty nodes  $V^c$ , with  $V = V^n \cup V^p \cup V^c$ . Then, we give the following definition:

*Definition 3.2.* A general topology structure in this article indicates that: Given a multi-agent system and a set of nodes  $V = V^n \cup V^p \cup V^c$ , then  $G[V^n]$  is fixed and rooted, while  $G - G[V^n]$  is arbitrary.

#### 3.2 System Model

In this article, we consider the following discrete-time dynamics model:

$$x_i(k+1) = x_i(k) + u_i(k), \quad i \in V, \quad (1)$$

with system initial state  $\mathbf{x}(0) = [x_1(0), \dots, x_n(0)]^T$  arbitrarily specified. In the following, the definition of different nodes will be specified in detail.

(i) **Normal Node:** A normal node updates its control input by

$$u_i(k) = \sum_{j \in N_i} \alpha_{ij}(k) \bar{x}_j(k) + \omega_i(k), \quad i \in V^n. \quad (2)$$

In the equation,  $\omega_i(k) \in \mathbb{R}$  represents bounded noise ( $|\omega_i(k)| < \omega$ ) induced by environment.  $\alpha_{ij}(k) (\geq 0)$  is called the adjacent weight from node  $j$  to node  $i$  satisfying  $\sum_{j \in N_i} \alpha_{ij}(k) < 1$ .

(ii) **Persistent Faulty Node (PFN):** A PFN exports a random value with a specified percentage  $p_p\%$  at each time as

$$u_i(k) = \text{Random} - x_i(k),^2 \quad i \in V^p, \quad (3)$$

and acts normal with averaged adjacent weights otherwise, i.e.,  $\alpha_{ij}(k) = \frac{1}{|N_i|+1}$  for  $j \in N_i$ . Random is a random variable that has a specified probability density function  $f_{\text{Random}}$ .

(iii) **Constant Faulty Node (CFN):** A CFN holds its initial value as the input with a specified percentage  $p_c\%$  by

$$u_i(k) = 0, \quad i \in V^c. \quad (4)$$

In other words, with  $1 - p_c\%$ , it acts as a normal node with averaged adjacent weights.

The reason that these three types of nodes are considered in our article is that they are indeed rather basic, which are also commonly seen in resilient systems [1, 13]. Meanwhile, in practical situations, the faulty nodes may attempt to remain hidden from time to time while carrying out its actions to avoid detection and decrease the risk of being identified [21, 23], and thus abnormal behavior with a specified percentage is considered in this article. Among these different nodes, we consider two kinds of attackers.

*Definition 3.3.* An *honest-but-curious attacker* is a normal node who follows all protocol steps correctly but is curious in an attempt to estimate the initial states of other nodes. A *malicious attacker* is a faulty node who intends to disrupt the consensus goal.

Following, we also give a definition of privacy preservation.

*Definition 3.4.* In a given network, the privacy of the initial value  $x_i(0)$  of a normal node  $i$  is preserved if neither honest-but-curious attackers nor malicious attackers can estimate its actual state.

Therefore, the goal in this article is to design proper control input  $u_i(k)$  for each normal node  $i$  in a distributed manner, so all normal agents achieve MAPPRC in a general topology structure, i.e., we aim to achieve

$$\limsup_{k \rightarrow \infty} \max_{i, j \in V^n} |x_i(k) - x_j(k)| < \varepsilon \quad (5)$$

for a sufficiently small  $\varepsilon$ , in the presence of faulty nodes and without the access of each node's actual state to its neighboring nodes.

*Remark 3.1.* It deserves to be mentioned that the detailed privacy index, such as with any guaranteed accuracy [32], or  $(\epsilon, \delta)$ -data-privacy [11], is related to the design of Laplace noise or the cryptography process, which is beyond the scope of this article.

Next, we will introduce the noise-based approach and the cryptography-based approach, respectively, to achieve MAPPRC through adjusting the adjacent weights in a distributed manner.

<sup>2</sup>It is also referred to as  $x_i(k+1) = \text{Random}$ .

#### 4 NOISE-BASED MAPPRC

In this section, we first introduce noise on agent state to achieve MAPPRC. In this instance, for each agent  $i$ ,

$$\bar{x}_i(k) = x_i(k) + v_i(k), \quad \forall i \in V, \quad (6)$$

where the specified noise  $v_i(k) \sim \text{Lap}(b_i(k))$  follows independent zero-mean random variable with Laplace distribution and variance  $2b^2$ , and with probability density function  $\mathcal{L}(x; b) = \frac{1}{2b} e^{-\frac{|x|}{b}}$  [7]. In the equation,

$$b_i(k) = cq^k \quad (7)$$

with  $c > 0, \frac{1}{2} < q < 1$ .

Similar to the original version, the modified Q-consensus algorithm is improved in its reward function and credibility function construction. Therefore, we introduce the current version directly. When agent  $i$  obtains information  $\bar{x}_j(k)$  from its neighbors for  $j \in N_i$ , an *immediate reward*  $r_{ij}$  is obtained by the relative distance  $\bar{x}_j - x_i$

$$r_{ij}(k) = \alpha_{ij}(k-1)f(|\bar{x}_j(k) - x_i(k) + \omega_{ij}(k)|), \quad \forall j \in N_i$$

where  $\omega_{ij}$  ( $|\omega_{ij}| \leq \omega$ ) is a bounded noise on the transmission channel from  $j$  to  $i$ . Remarkably, we introduce the adjacent weight  $\alpha_{ij}$  of the last step to take the previous impact into consideration so reward function becomes more reasonable. In the spirit of the idea that each node is in accordance with its own state, naturally, the reward function is defined inversely proportional to the relative distance  $|\bar{x}_j - x_j + \omega_{ij}|$ , e.g.,

$$f(|\bar{x}_j(k) - x_i(k) + \omega_{ij}(k)|) = e^{-|\bar{x}_j(k) - x_i(k) + \omega_{ij}(k)|\beta(k)}$$

with an appropriately designed parameter  $\beta(k) > 0$ . By the definition,  $0 < r_{ij} \leq 1$ . When the two states are close,  $r_{ij}$  is closer to 1; and the value tends to 0 otherwise.

Following, the *credibility*  $Q_{ij}$  is defined based on the immediate reward. We apply the implementation equation via the associated normalized reward by

$$Q_{ij}(k) = Q_{ij}(k-1) + \eta(k) \left( \frac{r_{ij}(k)}{\sum_{l \in N_i} r_{il}(k)} - Q_{ij}(k-1) \right),$$

$$Q_{ij}(0) = 1, \quad \forall j \in N_i,$$

where  $\eta(k)$ , as a step size, monotonously decreases to 0, with detailed setting given in the simulation section. Here, we utilize  $\frac{r_{ij}}{\sum_{l \in N_i} r_{il}}$  to replace of  $r_{ij}$  in previous version, so horizontal assessments are introduced and thus the credibility function is more rational accordingly.

Finally, the adjacent weight  $\alpha_{ij}$  is calculated. The basic rule is that for neighbors with a higher credibility, the adjacent weights are larger. Therefore, by normalization operation of the credibility for all neighboring nodes, we have

$$\alpha_{ij}(k) = \frac{Q_{ij}(k)}{\sum_{l \in N_i} Q_{il}(k)} \left( 1 - \frac{1}{|N_i| + 1} \right), \quad \forall j \in N_i,$$

where  $|N_i|$  is the cardinality of  $N_i$ . The design obviously satisfies  $\sum_{j \in N_i} \alpha_{ij}(k) = 1 - \frac{1}{|N_i| + 1}$ , and  $\alpha_{ii}(k) = \frac{1}{|N_i| + 1}$  for all  $k$ . The detailed algorithm is named Noise-based MAPPRC algorithm and summarized in Algorithm 1.

**ALGORITHM 1:** Noise-based MAPPRC for a normal node  $i$ 


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1 Initialize  $\alpha_{ii} = \alpha_{ij}(0) = \frac{1}{|N_i|+1}, Q_{ij}(0) = 1, j \in N_i$ 
2 for  $k = 1$  to  $T$  do
3   for  $j \in N_i$  do
4      $\bar{x}_j(k) = x_j(k) + v_j(k),$ 
5      $r_{ij}(k) = \alpha_{ij}(k-1)f(|\bar{x}_j(k) - x_i(k) + \omega_{ij}(k)|);$ 
6   end
7   for  $j \in N_i$  do
8      $Q_{ij}(k) = Q_{ij}(k-1) + \eta(k) \left( \frac{r_{ij}(k)}{\sum_{l \in N_i} r_{il}(k)} - Q_{ij}(k-1) \right);$ 
9   end
10  for  $j \in N_i$  do
11     $\alpha_{ij}(k) = \frac{Q_{ij}(k)}{\sum_{l \in N_i} Q_{il}(k)} \left( 1 - \frac{1}{|N_i|+1} \right);$ 
12  end
13   $x_i(k+1) = x_i(k) + \sum_{j \in N_i} \alpha_{ij}(k)(\bar{x}_j(k) - x_i(k) + \omega_{ij}(k)) + \omega_i(k);$ 
14 end

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**5 CRYPTOGRAPHY-BASED MAPPRC**

Generally speaking, on the one hand, the system with predefined noise involved will converge only if the noise tends to 0 and thus to involve the noise will result in lower convergence rate. On the other hand, accuracy of reward function will be affected by the introduced noise, which will lead to a decrease in success rate. To deal with the problems above, we apply Paillier cryptosystem in this section.

In the following, we will briefly review homomorphic encryption and confidential interaction protocol and then present our modified cryptography-based MAPPRC approach.

**5.1 Homomorphic Encryption**

Now, we introduce a cryptosystem to encrypt the node state without the assistance of a third party. In the system, both private key and public key are considered so any node could encrypt its state with a public key, while this information can only be decrypted by nodes that have access to the private key. Among all the available cryptosystems, we concentrate on a Paillier cryptosystem [31, 32] with the following basic functions:

- (1) Key generation:
  - (a) Choose two large prime numbers  $p \in \mathbb{Z}$  and  $q \in \mathbb{Z}$  of equal bit-length and denote  $\hat{n} = pq$ .
  - (b) Let  $\lambda = \phi(\hat{n}) = (p-1)(q-1)$  where  $\phi(\cdot)$  is the Euler's totient function.
  - (c) Let  $\mu = \phi(\hat{n})^{-1} \bmod \hat{n}$ , which is the modular multiplicative inverse of  $\phi(\hat{n})$ .
  - (d) The public key  $k_p$  is then  $\hat{n}$ .
  - (e) The private key  $k_s$  is then  $(\lambda, \mu)$ .
- (2) Encryption ( $c = \mathcal{E}(\hat{m}) \in \mathbb{Z}_{\hat{n}^2}^*$ ). Here, it is denoted that  $\mathbb{Z}_{\hat{n}} = \{z | z \in \mathbb{Z}, 0 \leq z < \hat{n}\}$  and  $\mathbb{Z}_{\hat{n}}^* = \{z | z \in \mathbb{Z}, 0 \leq z < \hat{n}, \gcd(z, \hat{n}) = 1\}$  where  $\gcd(a, b)$  is the greatest common divisor of  $a$  and  $b$ .
  - (a) Choose a random  $r \in \mathbb{Z}_{\hat{n}}^*$ .
  - (b) The ciphertext is given by  $c = (\hat{n} + 1)^{\hat{m}} r^{\hat{n}} \bmod \hat{n}^2$  where  $\hat{m} \in \mathbb{Z}_{\hat{n}}, c \in \mathbb{Z}_{\hat{n}^2}^*$ .
- (3) Decryption ( $\hat{m} = \mathcal{D}(c) \in \mathbb{Z}_{\hat{n}}$ ).
  - (a) Define the integer division function  $L(\mu) = \frac{\mu-1}{\hat{n}}$ .
  - (b) The plaintext is  $\hat{m} = L(c^{\lambda} \bmod \hat{n}^2) \mu \bmod \hat{n}$ .

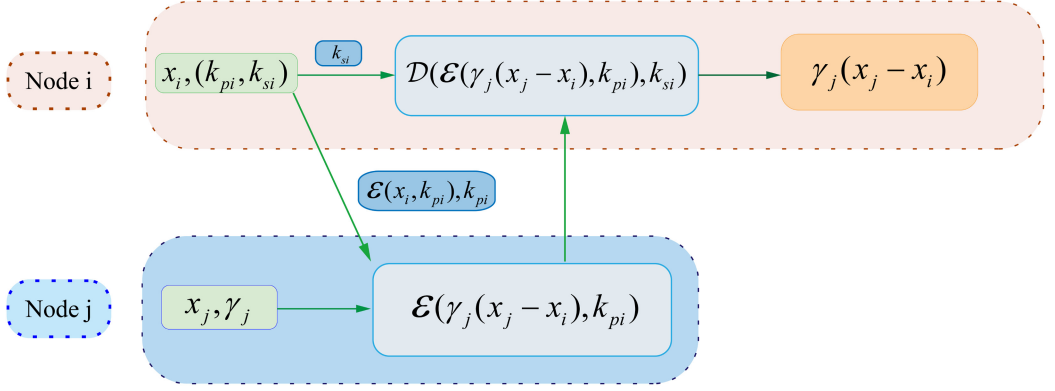


Fig. 3. The confidential interaction protocol of two neighboring nodes.

A cryptosystem is called homomorphic only if certain computation operations could be carried out on the encrypted ciphertext. In this way, Paillier cryptosystem belongs to homomorphic cryptosystem, since its ciphertext of  $\hat{m}_1 + \hat{m}_2$ , i.e.,  $\mathcal{E}(\hat{m}_1 + \hat{m}_2)$  could be obtained by  $\mathcal{E}(\hat{m}_1)$  and  $\mathcal{E}(\hat{m}_2)$  directly as follows:

$$\mathcal{E}(\hat{m}_1)\mathcal{E}(\hat{m}_2) = \mathcal{E}(\hat{m}_1 + \hat{m}_2).$$

## 5.2 Confidential Interaction Protocol

In the following, we take the process in which node  $i$  obtains the observed weighted state difference  $\gamma_j(x_j - x_i)$ , i.e.,  $\bar{x}_j$ , as an example to show how the state information is encrypted by Paillier cryptosystem and interacted between neighbors. First, node  $i$  sends its encrypted negative state  $\mathcal{E}_i(-x_i)$  as well as the public key  $k_{pi}$  to its neighbors. By utilizing  $k_{pi}$ , neighbor node  $j$  could obtain the encrypted state difference information  $\mathcal{E}_i(x_j - x_i)$ . After endowing  $\gamma_j$ -weighted difference in ciphertext  $\mathcal{E}_i(\gamma_j(x_j - x_i))$  and sending back to node  $i$ , the weighted state difference  $\gamma_j(x_j - x_i)$  is thus obtained. The detailed steps are given as follows:

- (1) Encrypt  $x_j$  with node  $i$ 's public key  $k_{pi}$  :  $x_j \rightarrow \mathcal{E}_i(x_j)$ .
- (2) Compute the difference directly in ciphertext:

$$\mathcal{E}_i(x_j - x_i) = \mathcal{E}_i(x_j + (-x_i)) = \mathcal{E}_i(x_j) \cdot \mathcal{E}_i(-x_i).$$

- (3) Compute the  $\gamma_j$ -weighted difference in ciphertext:

$$\mathcal{E}_i(\gamma_j(x_j - x_i)) = (\mathcal{E}_i(x_j - x_i))^{\gamma_j}.$$

Then, node  $j$  returns  $\mathcal{E}_i(\gamma_j(x_j - x_i))$  to node  $i$ . After receiving  $\mathcal{E}_i(\gamma_j(x_j - x_i))$ , node  $i$  decrypts it using the private key  $k_{si}$  to get the weighted difference:

$$\mathcal{E}_i(\gamma_j(x_j - x_i)) \xrightarrow{\mathcal{D}_i} \gamma_j(x_j - x_i).$$

The detailed process can be referred to in Reference [32], and the schematic diagram is shown in Figure 3. It deserves to be mentioned that information flows between neighboring agents in cryptography-based approach, and thus the topology here should be undirected.

## 5.3 Algorithm

Compared with the original version, the cryptography-based MAPPRC algorithm boasts rebuilt reward function with the previous impact taken into consideration and updated credibility function with horizontal assessments involved, which is similar to the noise-based approach. Besides,

it is conducted in two consecutive interactions with the reward function obtained under privacy-preserving requirement. In a word, the proposed approach involves a time horizon of length two, which consists of the following two layers of iteration:

The first layer is the decryption and state updating process. According to the Confidential Interaction Protocol, each node obtains the observed weighted state difference. The weighted coefficient for each node  $j$  at odd timestep is a bounded random value  $\gamma_j(2k-1)$ , and  $\frac{1}{\gamma_j(2k-1)}$  at even timestep. The significance of such construction is to eliminate the induced variable  $\gamma_j$  by multiplying two consecutive decrypted information. In addition, a variable  $\gamma_{max}$  satisfying  $\max\{\frac{1}{\gamma_{max}}\gamma_j(2k-1), \frac{1}{\gamma_{max}}\frac{1}{\gamma_j(2k-1)}\} < 1$  is introduced so the sum of all adjacent weights is less than 1. The second layer of iteration is to build the relation characterization for each pair of neighboring agents. Similar to noise-based MAPPRC approach, two concepts, *immediate reward*  $r_{ij}$  and *credibility*  $Q_{ij}$ , are defined subsequently. The detailed procedure is discussed as follows:

For node  $i$ , it first obtains two consecutive decrypted information  $\gamma_j(2k-1)(x_j(2k-2)-x_i(2k-2))$  and  $\frac{1}{\gamma_j(2k-1)}(x_j(2k-1)-x_i(2k-1))$  describing the state differences at dynamic instant, and an *immediate reward* is defined by the two inputs as

$$r_{ij}(2k) = \alpha_{ij}(2k-2)f(|x_j(2k-1)-x_i(2k-1)| \cdot |x_j(2k-2)-x_i(2k-2)|), \quad \forall j \in N_i.$$

In the same measure, the closer distance for two neighboring agents implies a larger reward value, and an example of function  $f$  is given as

$$\begin{aligned} & f(|x_j(2k-1)-x_i(2k-1)| \cdot |x_j(2k-2)-x_i(2k-2)|) \\ &= e^{-\sqrt{|x_j(2k-1)-x_i(2k-1)| \cdot |x_j(2k-2)-x_i(2k-2)|}\beta(k)}. \end{aligned}$$

Followed by this, the *credibility* is thus recursively updated by

$$Q_{ij}(2k) = Q_{ij}(2k-2) + \eta(2k) \cdot \left( \frac{r_{ij}(2k)}{\sum_{l \in N_i} r_{il}(2k)} - Q_{ij}(2k-2) \right), \quad \forall j \in N_i$$

with initial value  $Q_{ij}(0) = 1$ . The adjacent weight is also calculated as

$$\alpha_{ij}(2k) = \frac{Q_{ij}(2k)}{\sum_{l \in N_i} Q_{il}(2k)} \left( 1 - \frac{1}{|N_i| + 1} \right), \quad \forall j \in N_i.$$

The detailed algorithm, so-called cryptography-based MAPPRC algorithm, is summarized in Algorithm 2 as follows.

## 6 EXPERIMENT AND ANALYSIS

To validate our algorithm on MAPPRC problem, we conduct experiments in three ways, i.e., numerical simulations, smart-car hardware platform experiments, and distributed learning experiments. In numerical simulations, the noise-based and cryptography-based algorithms are applied in a fixed directed topology and an undirected one, respectively, in which the existing MSR-type algorithms are unavailable, and after this, the impact of hyperparameters on the system convergence is qualitatively analyzed. In smart-car hardware platform experiments, resilient rendezvous problem is conducted, therefore to evaluate effectiveness of our algorithms in the real-world application. In distributed learning experiments, a distributed machine learning platform is employed to achieve resilient training. It turns out that its performance by the proposed algorithm improves significantly, even when there are multiple unreliable training workers. Multimedia, code, and data are contained in the supplementary material.

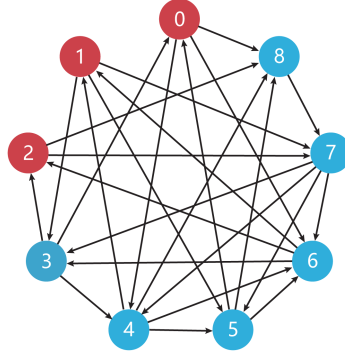


Fig. 4. A fixed directed network where 0, 1, and 2 are faulty nodes, and the rest normal nodes are rooted.

---

**ALGORITHM 2:** Cryptography-based MAPPRC for a normal node  $i$

---

```

1 Initialize  $\alpha_{ii} = \alpha_{ij}(0) = \frac{1}{|N_i|+1}$ ,  $Q_{ij}(0) = 1$ ,  $j \in N_i$ ;
2 for  $j \in N_i$  do
3   | Decrypt  $y_j(2k-1)(x_j(2k-2) - x_i(2k-2))$ ;
4 end
5  $x_i(2k-1) = x_i(2k-2) + \sum_{j \in N_i} \frac{1}{y_{\max}} \alpha_{ij}(2k-2) y_j(2k-1)(x_j(2k-2) - x_i(2k-2)) + \omega_i(2k-2)$ ;
6 for  $j \in N_i$  do
7   | Decrypt  $\frac{1}{y_j(2k-1)}(x_j(2k-1) - x_i(2k-1))$ ;
8   |  $r_{ij}(2k) = \alpha_{ij}(2k-2) f(|x_j(2k-1) - x_i(2k-1)| \cdot |x_j(2k-2) - x_i(2k-2)|)$ ;
9 end
10 for  $j \in N_i$  do
11   |  $Q_{ij}(2k) = Q_{ij}(2k-2) + \eta(2k) \left( \frac{r_{ij}(2k)}{\sum_{l \in N_i} r_{il}(2k)} - Q_{ij}(2k-2) \right)$ ;
12 end
13 for  $j \in N_i$  do
14   |  $\alpha_{ij}(2k) = \frac{Q_{ij}(2k)}{\sum_{l \in N_i} Q_{il}(2k)} \left( 1 - \frac{1}{|N_i|+1} \right)$ ;
15 end
16  $x_i(2k) = x_i(2k-1) + \sum_{j \in N_i} \frac{1}{y_{\max}} \alpha_{ij}(2k) \frac{1}{y_j(2k-1)} (x_j(2k-1) - x_i(2k-1)) + \omega_i(2k-1)$ ;

```

---

## 6.1 Numerical Simulations

**6.1.1 Experimental Details.** We first consider a fixed directed network with topology illustrated in Figure 4 (the topology turns to be undirected for cryptography-based case). Among all nine nodes (labeled by  $0, \dots, 8$ ) in the system, nodes 0, 1, and 2 are faulty nodes, and the rest normal ones are rooted. In the simulation setting, each of the agents has an arbitrarily initial state between 0 and 1. Apparently, neither the network connectivity nor network robustness is satisfied here, so the traditional MSR-type algorithms are not available. Two types of faulty nodes, PFN and CFN, are considered. The environment noise  $\omega_i(k)$  is a variable that follows Gaussian distribution:  $\omega_i(k) \sim N(\mu, \sigma^2)$ , where  $\mu = 0$ ,  $\sigma = \frac{1}{600}$ .

**6.1.2 Experimental Results.** After applying noise-based MAPPRC algorithm and cryptography-based MAPPRC algorithm on the given topology as Figure 4, the simulation results under different faulty nodes compositions and faulty probability are shown in Figures 5 and 6, respectively. Here,  $P$  is short for Persistent and  $C$  is short for Constant. In the simulation setting, we

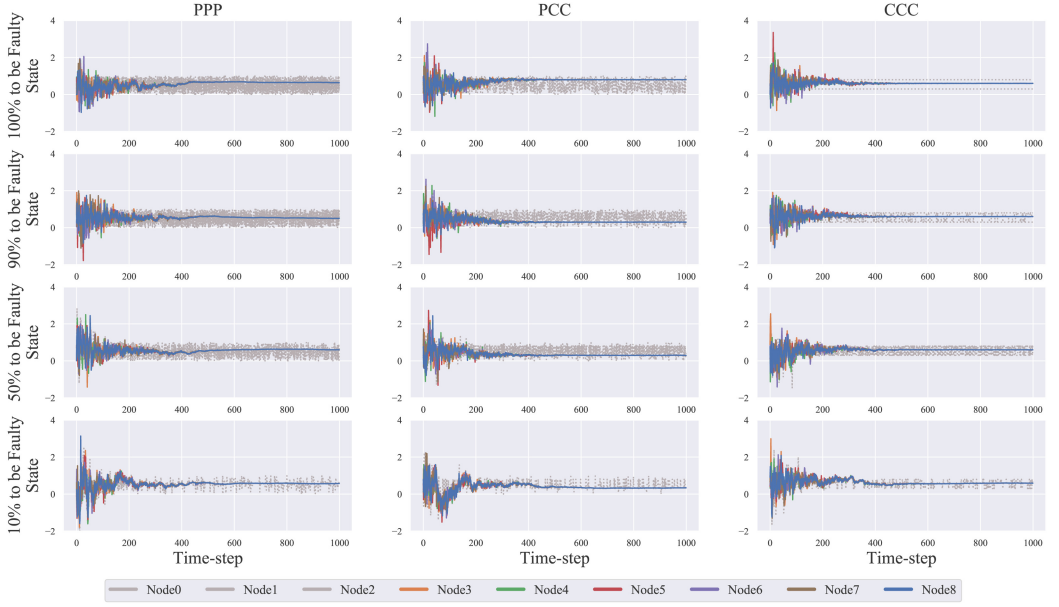


Fig. 5. The state evolution under noise-based MAPPRC algorithm.

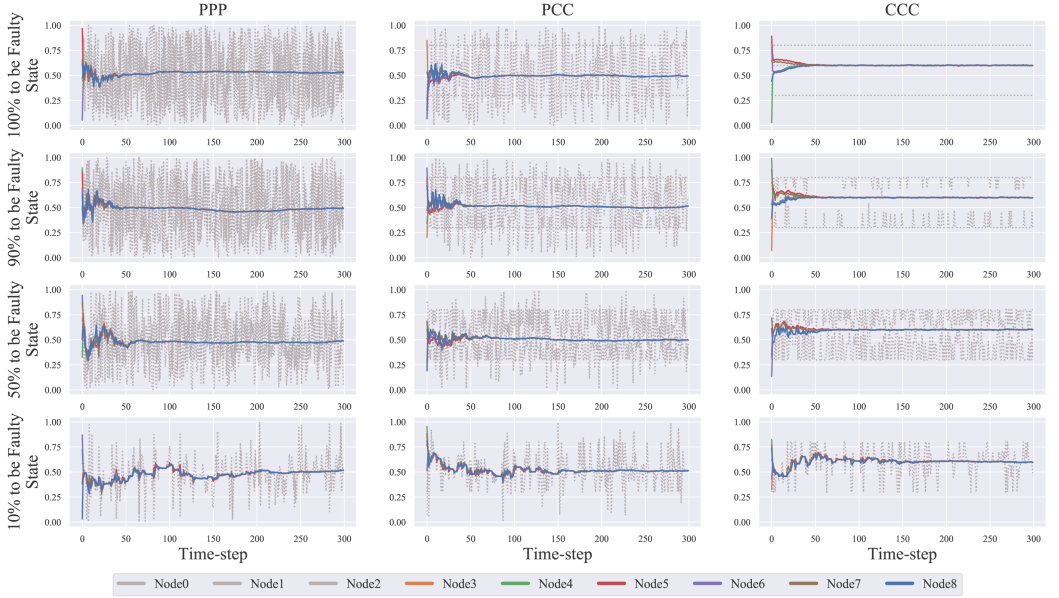


Fig. 6. The state evolution under cryptography-based MAPPRC algorithm.

let  $\beta(k) = 10 + 0.1k$ ,  $\eta(k) = 0.1$  if  $k < 0.8T$  else  $\eta(k+1) = \eta(k) - \eta(k)/0.2T$ , where  $T$  is the total evolution timestep. The induced noise  $v_i(k) \sim \text{Lap}(cq^k)$  is Laplace distributed with  $c = 1$ , and  $q = 0.99$ . The faulty node may update its state by following the faulty control law, i.e., Equation (3)/Equation (4), or by following the normal one, i.e., Equation (2). Formally, we define that the system reaches convergence only if  $x_{\max} - x_{\min} \leq \epsilon$  in successive 20 steps, where

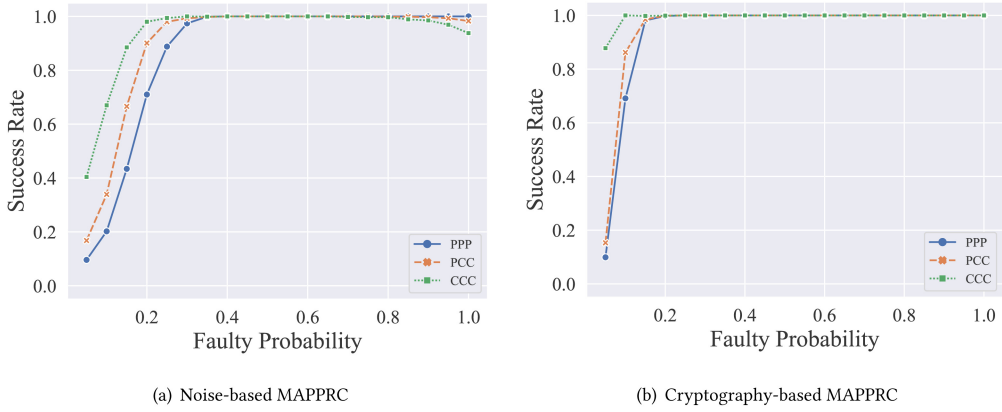


Fig. 7. The success rate with different faulty probability in (a) Noise-based MAPPRC algorithm and (b) Cryptography-based MAPPRC algorithm.

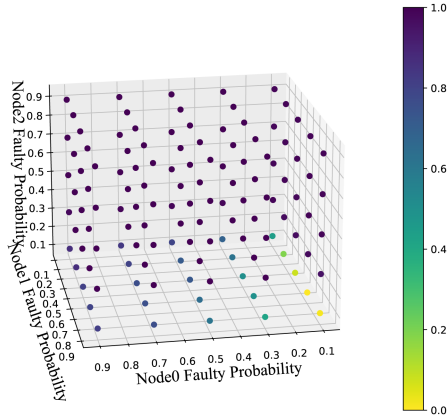


Fig. 8. The success rate with heterogenous faulty probabilities in cryptography-based MAPPRC algorithm.

$x_{max}/x_{min}$  denotes the maximum/minimum state among all normal nodes, and  $\epsilon$  is set 0.03 in our simulation. The results show that system can reach MAPPRC in all cases. Specifically, through the noise-based algorithm, normal nodes converge within 400 timesteps, while by cryptography-based algorithm, they converge within 150 timesteps.

Furthermore, we expect to analyze the relationship between faulty probability of faulty nodes and success rate. Through repeating experiments 1,000 times for different faulty nodes compositions PPP/PCC/CCC, the success rate with different faulty probability is given as Figure 7. From the results, it can be seen that within a small faulty probability region, specifically, (0, 0.3) for noise-based algorithm and (0, 0.2) for cryptography-based algorithm, the success rates increase sharply for both algorithms and then converge to 1. Generally speaking, the cryptography-based algorithm is of higher success rate compared to the noise-based algorithm, however, it suffers from communication and computation loads. We also consider a case of heterogenous faulty probabilities in cryptography-based MAPPRC algorithm, consisting of a faulty nodes composition PPP. By repeating experiments 1,000 times, the corresponding relationship between faulty probability of faulty nodes and success rate is shown in Figure 8, in which a darker color indicates a higher success rate. From the result, it can be seen that consensus success rate depends approximately on the

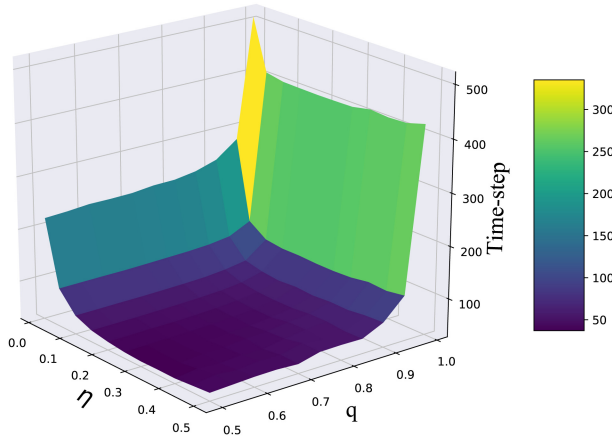


Fig. 9. The convergence timestep varies with step size  $\eta$  and Laplace noise parameter  $q$  in noise-based MAPPRC algorithm.

minimal faulty probability of faulty nodes, consistent with the above homogeneous experimental results. In particular, compared with the other two faulty nodes, node 2 has a greater impact on the convergence success rate. This may be determined by topology, as node 2 influences two other normal nodes, while both node 0 and node 1 influence three. The less information the normal nodes can obtain from the faulty node, the more difficult it can be identified.

Moreover, we provide an experiment to indicate how the convergence timestep varies with step size  $\eta$  and Laplace noise parameter  $q$ . The result shown in Figure 9 implies that convergence timestep is influenced by both step size  $\eta$  and Laplace noise parameter  $q$ . Specifically, when  $k < 0.8T$ ,  $\eta$  is fixed and the convergence timestep increases as  $q$  rises; when  $q$  is fixed, as  $\eta$  rises, the convergence timestep decreases.

Last, we compare the proposed modified version with original Q-consensus algorithm from the perspective of convergence rate and system stability. A network of nine agents is considered, with faulty agent numbers 1, 2, 3, respectively. We repeat the experiment 1,000 times, each of which satisfies the requirement of a general topology structure. Specifically, this process is realized by random connection among nodes and the cases not satisfying topology requirement are removed. The result in Figure 10 shows that the convergence rates of cryptography-based approach in all cases are faster compared with Q-consensus algorithm. In addition, to evaluate the stability of the system, two evaluation indexes, variance's average value and average value's variance, are introduced after system converges. Specifically, variance's average value indicates the average value of variances for all normal nodes in successive 100 steps, and average value's variance represents the variance of average values for all normal nodes in successive 100 steps. The detailed comparison is shown in Table 2 implying greater stability of the proposed MAPPRC algorithms.

**6.1.3 Large Scale Network.** In addition, we apply the proposed schemes in a large scale network composed of 900 normal nodes and 100 PFNs. The topology is randomly produced satisfying the assumption of at least one normal neighboring node, and all the rest conditions such as initial value and PFN performance are the same as small network. The simulation results under noise-based MAPPRC algorithm and cryptography-based MAPPRC algorithm are shown in Figure 11 verifying the effectiveness and fast convergence in large scale system as well.

In both noise-based MAPPRC algorithm and cryptography-based MAPPRC algorithm, for each normal node  $i$  in each iteration, the computational complexity is  $O(|N_i|T)$  and the space complexity

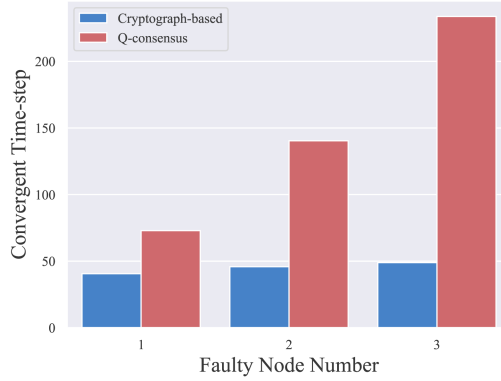


Fig. 10. The convergence timestep comparison of Q-consensus algorithm and cryptography-based MAPPRC algorithm.

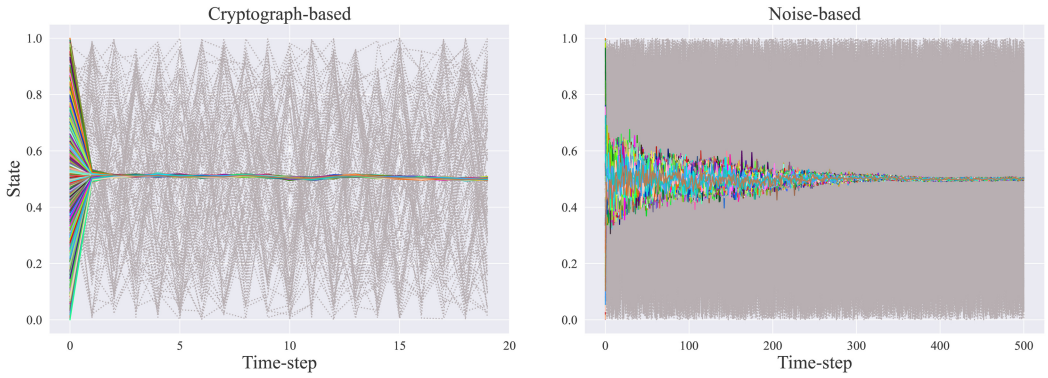


Fig. 11. The state evolution in a large-scale network with 900 normal nodes and 100 PFNs.

Table 2. Comparisons of Q-consensus Algorithm and MAPPRC

| Evaluation Index         | Q-consensus    | Noise-based MAPPRC | Cryptography-based MAPPRC |
|--------------------------|----------------|--------------------|---------------------------|
| Variance's average value | 1.42055788E-05 | 4.08398774E-06     | 2.63544542E-06            |
| Average value's variance | 2.54082552E-04 | 2.93346050E-05     | 3.71719805E-05            |

is  $O(|N_i|)$ , where  $T$  is the total timestep and  $|N_i|$  is the cardinality of  $N_i$ . Specifically, each normal node  $i$  needs  $O(|N_i|)$  to store the Q values and the weights of their neighbor nodes.

## 6.2 Resilient Rendezvous

In this subsection, we apply the proposed cryptography-based MAPPRC algorithm in a smart-car hardware platform to achieve resilient rendezvous control. In this experiment, the state space is nine square meters, i.e.,  $[0, 3] \times [0, 3]$ . Each vehicle is equipped with Mecanum wheels so it can move omnidirectionally. Meanwhile, a central computer is connected with OptiTrack that gathers information from all the vehicles involved, and each vehicle receives and utilizes information of its own and the neighbors' from the computer in a distributed manner. The platform consists of four vehicles labeled by 0, 1, 2, 3, with topology given as Figure 12. Among the vehicles, node 0 is stationary and node 1 is a PFN, while the other two are normal ones. The goal here is to identify and omit the disturbance of the faulty vehicles and to achieve resilient rendezvous control

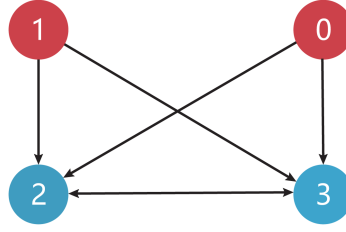


Fig. 12. The communication topology in the smart-car hardware platform.

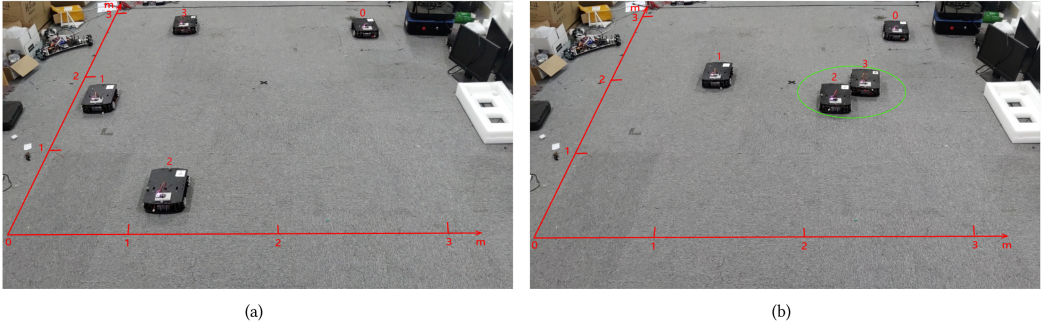


Fig. 13. (a) The initial states of the vehicles. (b) The normal vehicles (labeled by 2 and 3) rendezvous in the green circle.

with the normal ones. From the results, the initial locations of all vehicles are randomly produced, and normal vehicles achieve rendezvous gradually, with one snapshot of the experiment given as Figure 13. The corresponding trajectory of all vehicles is demonstrated as Figure 14. It is noted that to avoid collision in the actual situation, an offset (relative distance between green line and red line in the figure) is set on vehicles' states.<sup>3</sup>

### 6.3 Distributed Resilient Learning

Last, we apply our approach in image recognition on MNIST datasets [2, 19] and compare it with vanilla distributed learning algorithm.

**6.3.1 Experimental Details.** The framework of the distributed learning consists of a single parameter server and multiple workers. The training data is partitioned into different workers, and these workers independently compute the gradients based on their own data. The server aggregates these gradients and further uses the aggregated gradient to update the parameters of the system model. At intervals, the updated parameters are synchronized to the workers. In classic **Gradient Aggregation Rule (GAR)** [22], the adjacent weights from the workers are commonly identical. However, on account of the external attacks or internal crash, inevitably, the server may receive incorrect gradients, which will result in training task failure. Therefore, we aim to adjust the adjacent weights via the modified Q-consensus algorithm to eliminate the negative influence of incorrect gradients.

In the distributed learning experiments, first, we give the *safety assumption* as follows:

- The parameter server and at least one worker are reliable.
- The data on workers and server is independent and identically distributed (i.i.d.).

<sup>3</sup>The video link is <https://drive.google.com/file/d/1YaM65eQ0au6lpjNlmygwDY8wzJAMrjCP/view?usp=sharing>.

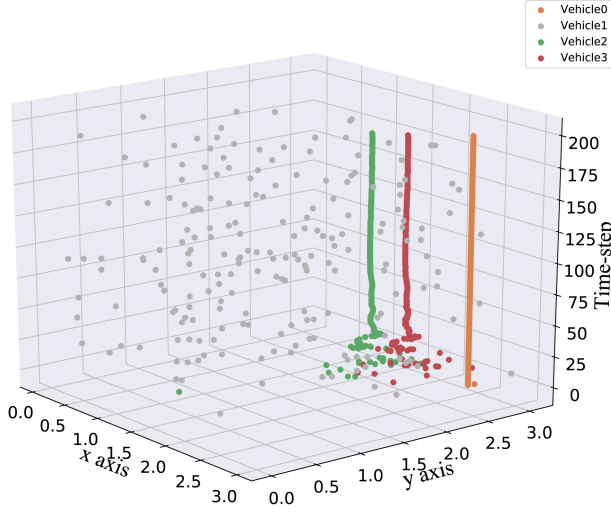


Fig. 14. The trajectory of the four vehicles.

In this experiment, server's state  $X_s(k)$  implies its gradient at timestep  $k$ , and  $N_s$  denotes the set of workers. Correspondingly,  $X_i(k)$ ,  $i \in N_s$  is the gradient of worker  $i$ . First, each worker  $i$  copies the model parameter  $\theta_s(k-1)$  from the server, to obtain its current gradient  $X_i(k)$ . Following, the server obtains its reward function of worker  $i$  by

$$r_i(k) = \alpha_i(k-1)f(|X_i(k) - X_s(k)|), \quad (8)$$

the credibility function by

$$Q_i(k) = Q_i(k-1) + \eta(k) \left( \frac{r_i(k)}{\sum_{l \in N_s} r_l(k)} - Q_i(k-1) \right),$$

and then normalizes the corresponding adjacent weight by

$$\alpha_i(k) = \frac{Q_i(k)}{\sum_{l \in N_s} Q_l(k)} \left( 1 - \frac{1}{|N_s| + 1} \right).$$

Finally, the model parameter  $\theta_s(k)$  of the server is calculated by aggregated gradients as

$$\theta_s(k) = \theta_s(k-1) - \lambda \left( \alpha_s X_s(k) + \sum_{i \in N_s} \alpha_i(k) X_i(k) \right),$$

where  $\lambda \in (0, 1]$  is the learning rate. It is noted that model parameter  $\theta$  contains about 25k weights in this experiment, so does the dimension of gradient  $X$ . Generally speaking, it is inefficient and also unnecessary to calculate the reward value, i.e., Equation (8), by utilizing the entire gradient dimensions. Therefore, only a hundred dimensions are randomly sampled during each iteration, to avoid the redundant computation and to accelerate the algorithm. The detailed algorithm is summarized in Algorithm 3 in Appendix A.

**6.3.2 Experimental Results.** In this distributed learning experiment, we consider one server and 10 workers, in which the faulty workers behave as PFNs. The system model is, essentially, a neural network consisting of an input layer, a hidden layer, and an output layer that use the ReLU activation function. The batch size is 256, and the learning rate  $\lambda$  is set 0.01. In addition, we use CGAR/CGAR2/CGAR5/CGAR7, and OURS/OURS2/OURS5/OURS7 to represent 0/2/5/7 faulty

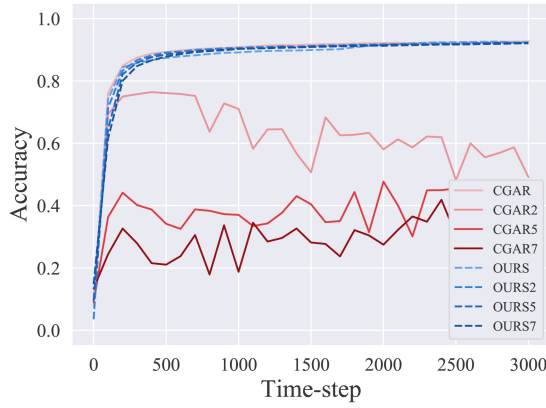


Fig. 15. The training process of different algorithms.

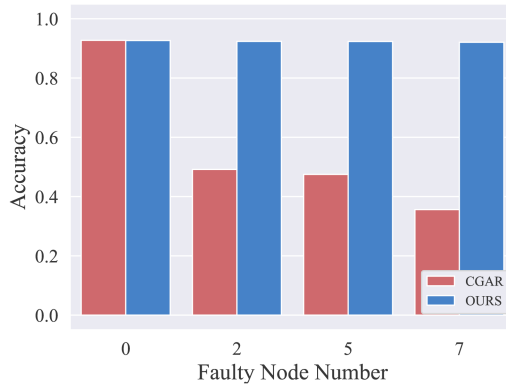


Fig. 16. The test accuracy of different algorithms.

workers in classic GAR algorithm and in our algorithm, respectively, with the training process given as Figure 15. At last, from the test accuracy of the two algorithms shown in Figure 16, we could obtain that (1) When system contains 0/2/5/7 faulty workers, classic GAR algorithm and our algorithm can achieve test accuracy rate of 92.72%/ 49.15%/ 47.48%/35.60%, and 92.66%/ 92.37%/ 92.33%/ 92.07%, respectively. (2) Compared with classic GAR, our algorithm shows improvement on test accuracy by 87.9%/94.5%/158.6% when system contains 2/5/7 faulty workers.

## 7 CONCLUSIONS

Taking both resilience and privacy-preserving requirements in multi-agent consensus into consideration, this article adopted two approaches, i.e., noise-based MAPPRC and cryptography-based MAPPRC algorithms to conduct experiments, respectively, by introducing specified noise function and Paillier cryptosystem on state. Meanwhile, both reward function and credibility function are reconstructed so the algorithm's convergence rate and stability are improved. Following, experiments in three ways, i.e., numerical simulations, resilient rendezvous of multiple smart cars, and privacy-preserving distributed resilient learning, were conducted to validate the effectiveness of the proposed algorithms. In spite of system noise, the experimental results illustrated that the proposed algorithms can achieve MAPPRC in any general topology structure. The success of

noise-based MAPPRC/cryptography-based MAPPRC algorithms in this area demonstrates their great potential for medical applications.

With a fixed topology without time-delay and two kinds of faulty nodes taken into account, this article shows positive results about our topic in a more ideal way. However, due to the complex dynamic environment, the topology is commonly time-varying and the behaviors of nodes are different in practical applications. Therefore, works with more dynamic topology conditions and various faulty nodes are still needed.

Besides, the success of the proposed algorithms on the numerical simulations, resilient rendezvous, and distributed resilient learning show the potential to be applied to more real-world scenarios, such as disease diagnosis, as we mentioned before. In the future, we aim to apply the proposed algorithms in medical scenarios to facilitate research and analysis while maintaining patients' privacy.

## APPENDIX

### A MAPPRC FOR DISTRIBUTED RESILIENT LEARNING

We modify the MAPPRC algorithm designed in Sections 4 and 5 and apply the modified algorithm to distributed resilient learning. The detailed algorithm is summarized in Algorithm 3.

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#### ALGORITHM 3: MAPPRC for distributed resilient learning

---

```

1 Initialize  $\alpha_s = \alpha_i(0) = \frac{1}{|N_s|+1}$ ,  $Q_i(0) = 1$ ,  $i \in N_s$ ;
2 Initialize the model parameter  $\theta_s(0)$  randomly;
3 for  $k = 1$  to  $T$  do
4   for  $i \in N_s$  do
5     Copy  $\theta_s(k-1)$  to  $\theta_i(k-1)$ ;
6     Calculate its gradient  $X_i(k)$  and send to the server;
7   end
8   for  $i \in N_s$  do
9      $r_i(k) = \alpha_i(k-1)f(|X_i(k) - X_s(k)|)$ ;
10  end
11  for  $i \in N_s$  do
12     $Q_i(k) = Q_i(k-1) + \eta(k) \left( \frac{r_i(k)}{\sum_{l \in N_s} r_l(k)} - Q_i(k-1) \right)$ ;
13  end
14  for  $i \in N_s$  do
15     $\alpha_i(k) = \frac{Q_i(k)}{\sum_{l \in N_s} Q_l(k)} \left( 1 - \frac{1}{|N_s|+1} \right)$ ;
16  end
17   $\theta_s(k) = \theta_s(k-1) - \lambda \left( \alpha_s X_s(k) + \sum_{i \in N_s} \alpha_i(k) X_i(k) \right)$ .
18 end
```

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## REFERENCES

- [1] J. S. Baras and X. Liu. 2019. Trust is the cure to distributed consensus with adversaries. In *Proceedings of the 27th Mediterranean Conference on Control and Automation (MED)*. 195–202.
- [2] L. Deng. 2012. The MNIST database of handwritten digit images for machine learning research. *IEEE Sig. Process. Mag.* 29, 6 (2012), 141–142.
- [3] S. M. Dibaji and H. Ishii. 2017. Resilient consensus of second-order agent networks: Asynchronous update rules over robust graphs. *Automatica* 81 (2017), 123–132.
- [4] H. Du, W. Zhu, G. Wen, Z. Duan, and J. Lv. 2017. Distributed formation control of multiple quadrotor aircraft based on nonsmooth consensus algorithms. *IEEE Trans. Cyber.* 49, 1 (2017), 342–353.

- [5] W. Fang, M. Zamani, and Z. Chen. 2021. Secure and privacy preserving consensus for second-order systems based on paillier encryption. *Syst. Contr. Lett.* 148 (2021), 104869.
- [6] D. Fiore and G. Russo. 2019. Resilient and private consensus in multi-agent systems. In *Proceedings of the European Control Conference*. IEEE, 3478–3483.
- [7] D. Fiore and G. Russo. 2019. Resilient consensus for multi-agent systems subject to differential privacy requirements. *Automatica* 106 (2019), 18–26.
- [8] C. N. Hadjicostis and A. D. Domínguez-García. 2020. Privacy-preserving distributed averaging via homomorphically encrypted ratio consensus. *IEEE Trans. Automat. Contr.* 65, 9 (2020), 3887–3894.
- [9] A. Haseltalab and M. Akar. 2015. Approximate byzantine consensus in faulty asynchronous networks. In *Proceedings of the American Control Conference*. IEEE, 1591–1596.
- [10] A. Haseltalab and M. Akar. 2015. Convergence rate analysis of a fault-tolerant distributed consensus algorithm. In *Proceedings of the Conference on Decision and Control*. IEEE, 5111–5116.
- [11] J. He, L. Cai, P. Cheng, J. Pan, and L. Shi. 2019. Consensus-based data-privacy preserving data aggregation. *IEEE Trans. Automat. Contr.* 64, 12 (2019), 5222–5229.
- [12] J. He, L. Cai, C. Zhao, P. Cheng, and X. Guan. 2019. Privacy-preserving average consensus: Privacy analysis and algorithm design. *IEEE Trans. Sig. Inf. Process. Netw.* 5, 1 (2019), 127–138.
- [13] J. Hou, F. Wang, L. Wang, and Z. Chen. 2021. Reinforcement learning based multi-agent resilient control: From deep neural networks to an adaptive law. In *Proceedings of the AAAI Conference on Artificial Intelligence*. IEEE, 7737–7745.
- [14] H. Ishii and A. Suzuki. 2018. Distributed randomized algorithms for PageRank computation: Recent advances. Uncertainty in complex networked systems. In *Honor of Roberto Tempo* (2018), 419–447.
- [15] V. Katewa, F. Pasqualetti, and V. Gupta. 2018. On privacy vs. cooperation in multi-agent systems. *Int. J. Contr.* 91, 7 (2018), 1693–1707.
- [16] R. Lazzaretto, S. Horn, P. Braca, and P. Willett. 2014. Secure multi-party consensus gossip algorithms. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 7406–7410.
- [17] H. J. LeBlanc and X. D. Koutsoukos. 2011. Consensus in networked multi-agent systems with adversaries. In *Proceedings of the 14th International Conference on Hybrid Systems: Computation and Control*. 281–290.
- [18] H. J. LeBlanc and X. D. Koutsoukos. 2012. Low complexity resilient consensus in networked multi-agent systems with adversaries. In *Proceedings of the 15th ACM International Conference on Hybrid Systems: Computation and Control*. 5–14.
- [19] Y. LeCun, L. Bottou, Y. Bengio, and P. Haffner. 1998. Gradient-based learning applied to document recognition. *Proc. IEEE* 86, 11 (1998), 2278–2324.
- [20] Nianyu Li, Christos Tsigkanos, Zhi Jin, Schahram Dustdar, Zhenjiang Hu, and Carlo Ghezzi. 2019. POET: Privacy on the edge with bidirectional data transformations. In *Proceedings of the IEEE International Conference on Pervasive Computing and Communications*. IEEE, 1–10.
- [21] N. Li, M. Zhang, E. Kang, and D. Garlan. 2021. Engineering secure self-adaptive systems with Bayesian games. *Fundam. Approach. Softw. Eng.* 12649 (2021), 130–151.
- [22] X. Lian, Y. Huang, Y. Li, and J. Liu. 2015. Asynchronous parallel stochastic gradient for nonconvex optimization. In *Advances in Neural Information Processing Systems* 28. IEEE, Montreal, Quebec, Canada, 2737–2745.
- [23] E. T. Mahsa. 2017. A game-theoretic decision-making framework for engineering self-protecting software systems. In *Proceedings of the IEEE/ACM International Conference on Software Engineering Companion*. IEEE, 449–452.
- [24] N. E. Manitara and C. N. Hadjicostis. 2013. Privacy-preserving asymptotic average consensus. In *Proceedings of the European Control Conference*. IEEE, 760–765.
- [25] D. G. Mikulski, F. L. Lewis, E. Y. Gu, and G. R. Hudas. 2012. Trust method for multi-agent consensus. In *Proceedings of the Conference on Unmanned Systems Technology*. 1–15.
- [26] Y. Mo and R. M. Murray. 2016. Privacy preserving average consensus. *IEEE Trans. Automat. Control* 62, 2 (2016), 753–765.
- [27] Dinh C. Nguyen, Ming Ding, Pubudu N. Pathirana, Aruna Seneviratne, and Albert Y. Zomaya. 2022. Federated learning for COVID-19 detection with generative adversarial networks in edge cloud computing. *IEEE Internet Things J.* 9, 12 (2022), 10257–10271.
- [28] B. Ning, Q. Han, and Z. Zuo. 2017. Distributed optimization for multiagent systems: An edge-based fixed-time consensus approach. *IEEE Trans. Cyber.* 49, 1 (2017), 122–132.
- [29] X. Pan, M. Zhang, D. Wu, Q. Xiao, S. Ji, and Z. Yang. 2020. Justinian’s GAAvernor: Robust distributed learning with gradient aggregation agent. In *Proceedings of the USENIX Security Symposium*. IEEE, 1641–1658.
- [30] G. Ramos, D. Silvestre, and C. Silvestre. 2022. General resilient consensus algorithms. *Internat. J. Control* 95, 6 (2022), 1482–1496.
- [31] M. Ruan, M. Ahmad, and Y. Wang. 2017. Secure and privacy-preserving average consensus. In *Proceedings of the Workshop on Security and Privacy in Cyber-Physical Systems*. IEEE, 123–129.

- [32] M. Ruan, H. Gao, and Y. Wang. 2019. Secure and privacy-preserving consensus. *IEEE Trans. Automat. Control* 64, 10 (2019), 4035–4049.
- [33] K. Saulnier, D. Saldana, A. Prorok, G. J. Pappas, and V. Kumar. 2017. Resilient flocking for mobile robot teams. *IEEE Robot. Automat. Lett.* 2, 2 (2017), 1039–1046.
- [34] D. M. Senejohnny, S. Sundaram, C. De Persis, and P. Tesi. 2019. Resilience against misbehaving nodes in asynchronous networks. *Automatica* 104 (2019), 26–33.
- [35] S. Sundaram and C. N. Hadjicostis. 2011. Distributed function calculation via linear iterative strategies in the presence of malicious agents. *IEEE Trans. Automat. Contr.* 56, 7 (2011), 1495–1508.
- [36] Z. Yang and W. U. Bajwa. 2019. ByRDIE: Byzantine-resilient distributed coordinate descent for decentralized learning. *IEEE Trans. Sig. Inf. Process. Netw.* 5, 4 (2019), 611–627.
- [37] H. Zhang and S. Sundaram. 2012. Robustness of information diffusion algorithms to locally bounded adversaries. In *Proceedings of the American Control Conference*. 5855–5861.
- [38] J. Zhang, J. Tan, and J. Wang. 2021. Privacy security in control systems. *Sci. China Inf. Sci.* 64 (2021), 176201.

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